Saif Shikalgar

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Education

G. H. Raisoni College of Engineering and Management — Pune, India
B.Tech in Computer Science & Engineering (Cybersecurity Specialization); ***CGPA: 8.10/10.0*** — *Sep 2023 – May 2027 (Expected)*

Coursework: Operating Systems, Computer Networks, Data Structures & Algorithms, Microcontrollers & Embedded Systems, Network Security & Cryptography, Object-Oriented Programming (OOP), DBMS.

Technical Skills

Languages: C, C++ (C11/C++20), Python, Kotlin, Java, JavaScript, TypeScript, Bash, SQL, Solidity.
Embedded & Hardware: STM32 (HAL/CMSIS), ESP32, FreeRTOS, ARM Cortex-M, CAN Bus 2.0B / TWAI, OBD-II, SPI, I2C, UART, PWM, BLE GATT, NVIDIA Jetson Nano/Orin, AS5600 Encoders.
Automotive & Security: Dual-Gate CAN IPS/IDS, Mahalanobis Anomaly Engine, ISO/SAE 21434 Concepts, Wireshark, Packet Fuzzing, Jitter Filtering, Exploit-DB, CVE Auditing, Penetration Testing.
Mobile & Full-Stack: Android SDK, Jetpack Compose, MVVM Architecture, React.js, Next.js 14, FastAPI, Node.js, Tailwind CSS, PostgreSQL, Firebase (Auth, Firestore), Google Maps / OSM SDK.
AI, Cloud & Tools: PyTorch, TorchVision, OpenAI CLIP, YOLOv8, FAISS Vector Search, CUDA, Docker, Git, GCP Cloud Functions, Linux (Ubuntu/Debian).

Work Experience & Engineering Leadership

AuraByte Studios (Independent Venture) — Pune, India
Lead Android App Developer & Founder — *Dec 2025 – Present*

- Designed, developed, and deployed **FiberOpticCalc** on Google Play Store, scaling to **5.73K+ installs, 2.09K+ active devices, and a 4.6⋆ rating** with a 45.0% conversion rate and recurring revenue.
- Architected native application using **Kotlin**, **Jetpack Compose**, and **MVVM** clean architecture, strictly adhering to SOLID design principles and eliminating runtime memory leaks.
- Engineered a multi-engine mapping system (**OpenStreetMap / Google Maps SDK**) with a graph-traversal **OTDR fault localization algorithm** to calculate fiber cable breaks to sub-meter street precision.
- Implemented an **Atomic Persistence Engine** preventing data corruption during offline operations and schema-aware cloud synchronization via the **Google Drive REST API**.
- Deployed serverless microservices on **GCP Cloud Functions** for automated Google Play Billing validation, webhook processing, and database state updates.

Team Abhyuday Racing — Pune, India
Technical Lead & Department Head – Drive-by-Wire (Automotive R&D) — *Jan 2025 – Present*

- Designed and deployed a distributed **3-ECU network (NVIDIA Jetson Nano + STM32F446RE + ESP32)** over CAN 2.0B bus with priority arbitration and 99.9% packet reliability under high EMI.
- Programmed real-time deterministic **Autonomous Emergency Braking (AEB)** control logic, securing **1st Place Nationally at aBAJA 2026** by halting safely at **6.2m** at 30 km/h with **40 bar hydraulic line pressure**.
- Implemented automotive-grade **failsafe supervision** with a **50 Hz CAN heartbeat loop**, 10 Hz telemetry scheduling, node-loss detection, and visual fault indicators, winning **1st Place in ACC** at aBAJA 2025.
- Engineered a modular Brake-by-Wire linear actuator system mounted parallel to the tandem master cylinder, cutting costs by **50%** and winning the **1st Place Manufacturing Excellence Award**.

Key Technical Projects

Dual-Gate STM32 Hardware CAN Firewall & Anomaly IDS/IPS | *STM32F446RE, C11, CMSIS/HAL, FreeRTOS, CAN 2.0B*

- Engineered an inline dual-transceiver CAN bus firewall intercepting OBD-II injection, fuzzing, and RPM spoofing attacks with a deterministic **5.92 μs processing latency** (1,065 clock cycles at 180 MHz).
- Implemented fixed-point Mahalanobis distance anomaly detection and jitter statistical analysis, rejecting **88.5% of noise** and achieving a **99%+ threat mitigation rate** with zero bus contention.

CYO: Local-First Multimodal AI Image Intelligence System | *PyTorch, CLIP, YOLOv8, FAISS, FastAPI, React, Docker, CUDA*

- Developed a privacy-first semantic search engine combining OpenAI CLIP embeddings (ViT-B/32) and YOLOv8 for natural-language text-to-image queries over local image galleries with zero cloud API dependency.
- Implemented a FAISS vector database achieving **sub-100ms retrieval latency** across 1,000+ high-res images; containerized via Docker with custom NVIDIA CUDA runtimes for bare-metal GPU acceleration.

Wireless Penetration Testing & 802.11 Security Toolkit | *ESP32, C++, 802.11 Protocols, Wireshark, PCAP*

- Engineered a portable 802.11 auditing tool capable of passive beacon sniffing, targeted deauthentication, and WPA2 EAPOL/PMKID handshake capture for PCAP analysis in Wireshark.

Suraksha360: Plug-and-Play IoT Security Auditor & CVE Scanner | *C++, ESP8266, Exploit-DB, Windows Registry API*

- Created a plug-and-play USB HID hardware security auditing tool executing zero-install system configuration analysis on Windows hosts, cross-referencing software against Exploit-DB for known CVE vulnerabilities.

Honors & Awards

1st Place, Autonomous Emergency Braking (AEB): National aBAJA 2026 Competition; achieved a 6.2m halt on a 6.0m autonomous target threshold at 30 km/h.
1st Place, Manufacturing Excellence Award: National aBAJA 2026 Competition; recognized for modular Drive-by-Wire and Brake-by-Wire actuator design.
1st Runner Up, MATLAB Advanced Simulation: National aBAJA 2026 Competition; developed a GPS-based Point-A to Point-B autonomous traversal algorithm.
1st Place, Adaptive Cruise Control (ACC): aBAJA 2025; achieved following an 8-hour high-stakes crisis hardware migration from STM32 to ESP32.